STATISTICS & MACHINE LEARNING · LESSON 2 OF 28

Probability and Distributions: Rules, Bayes, Binomial, Poisson and Normal

The language of uncertainty. Probability rules, conditional probability and Bayes' theorem (with a fraud-alert example), random variables and expected value, and the distributions analysts use most (Bernoulli, binomial, Poisson, uniform, normal, exponential), with scipy.stats and simulation.

Python class 14 | SciPy 1.1x · NumPy 2 | Beginner | 110 min

By the end of this lesson you can

- Apply the addition, multiplication and complement rules and conditional probability
- Use Bayes' theorem to update probabilities and explain base-rate effects
- Compute expected value and variance of a random variable
- Model counts and measurements with binomial, Poisson, normal and exponential distributions using scipy.stats
- Use the empirical (68-95-99.7) rule and check normality with a Q-Q plot

Datasets used

`delivery_times.csv` `ab_test.csv`

Probability basics

Probability is a number from 0 (impossible) to 1 (certain). For equally likely outcomes, P(event) = favourable outcomes ÷ all outcomes. In data, it is a long-run frequency.

Python

```
import numpy as np, pandas as pd
from scipy import stats
rng = np.random.default_rng(7)
rolls = rng.integers(1, 7, 100_000)
print("P(six) theory 0.1667 | simulated", round((rolls == 6).mean(), 4))
print("P(even or > 4) theory", round(4 / 6, 4), "| simulated", round(((rolls % 2 == 0) |
(rolls > 4)).mean(), 4))
```

OUTPUT

```
P(six) theory 0.1667 | simulated 0.1682
P(even or > 4) theory 0.6667 | simulated 0.6679
```

Rule	Formula	Example
Complement	P(not A) = 1 − P(A)	P(no conversion) = 1 − 0.10
Addition	P(A or B) = P(A) + P(B) − P(A and B)	Even or > 4: 3/6 + 2/6 − 1/6
Multiplication (independent)	P(A and B) = P(A) × P(B)	Two sixes: 1/36
Conditional	P(A | B) = P(A and B) ÷ P(B)	P(churn | month-to-month)

Python

```python
tc = pd.DataFrame({"contract": ["M2M"] * 1690 + ["1yr"] * 739 + ["2yr"] * 571,
                   "churn": [1] * 607 + [0] * 1083 + [1] * 125 + [0] * 614 + [1] * 40 + [0]
* 531})   # counts from the churn EDA
joint = pd.crosstab(tc.contract, tc.churn, normalize="all").round(3)
print(joint)
print("P(churn) =", round(tc.churn.mean(), 3), "| P(churn | M2M) =", round(tc[tc.contract
== "M2M"].churn.mean(), 3), "| P(M2M | churn) =", round((tc[tc.churn == 1].contract ==
"M2M").mean(), 3))
```

OUTPUT

```
churn         0      1
contract
1yr       0.205  0.042
2yr       0.177  0.013
M2M       0.361  0.202
P(churn) = 0.257 | P(churn | M2M) = 0.359 | P(M2M | churn) = 0.786
```

P(churn | M2M) and P(M2M | churn) are **different** questions. Mixing them up is one of the most common reasoning errors.

Bayes' theorem

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

A fraud model flags 99% of fraudulent payments and wrongly flags 2% of genuine ones. Only 0.5% of payments are fraud. If a payment is flagged, how likely is it to be fraud?

Python

```
p_fraud, sens, fpr = 0.005, 0.99, 0.02
p_flag = sens * p_fraud + fpr * (1 - p_fraud)                # law of total probability
p_fraud_given_flag = sens * p_fraud / p_flag
print(f"P(flag) = {p_flag:.4f} | P(fraud | flag) = {p_fraud_given_flag:.1%}")
n = 100_000; fraud = rng.random(n) < p_fraud
flag = np.where(fraud, rng.random(n) < sens, rng.random(n) < fpr)
print(f"simulated: {fraud[flag].mean():.1%} of flagged payments are fraud ({flag.sum():,} flags, {(fraud & flag).sum()} real fraud)")
```

OUTPUT

```
P(flag) = 0.0249 | P(fraud | flag) = 19.9%
simulated: 19.9% of flagged payments are fraud (2,485 flags, 495 real fraud)
```

Only about one flag in five is real fraud, because genuine payments vastly outnumber fraud (the **base rate**). This is why precision matters in imbalanced ML problems (lesson 21) and why naive Bayes classifiers work (lesson 20).

Random variables and expected value

A **random variable** assigns a number to each outcome. Its **expected value** is the probability-weighted average: the long-run mean.

Python

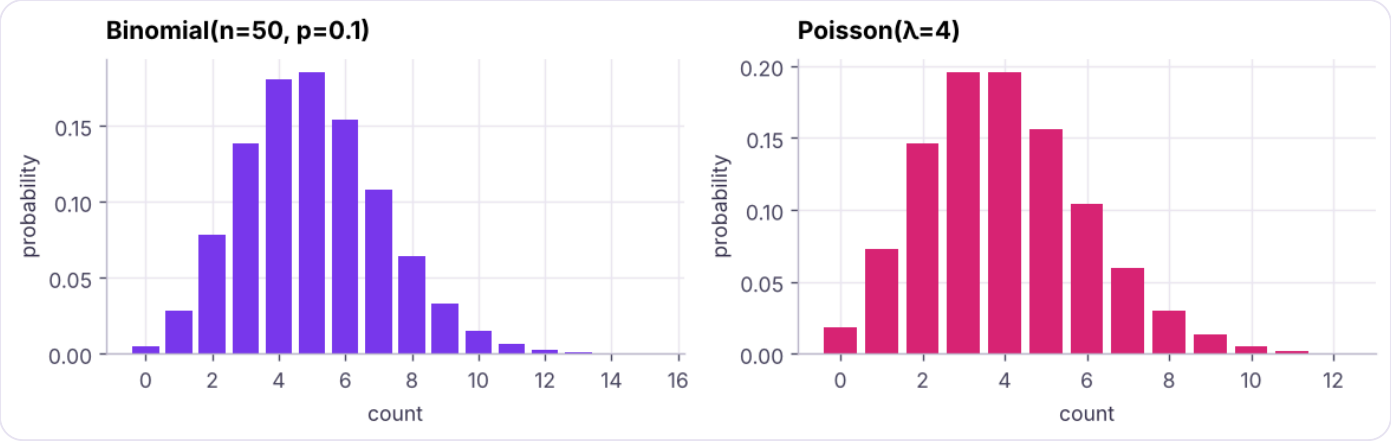
```
values = np.array([0, 500, 2000]); probs = np.array([0.80, 0.15, 0.05])      # customer spend next month
ev = (values * probs).sum(); var = ((values - ev) ** 2 * probs).sum()
print(f"E[spend] = ₹{ev:.0f} | sd = ₹{var ** 0.5:.0f}")
cost, uplift = 60, 0.25                                                      # campaign: costs ₹60, raises spend by 25%
print(f"expected profit per customer from the campaign: ₹{ev * uplift - cost:.0f} → {'run it' if ev * uplift > cost else 'do not run it'}")
```

OUTPUT

```
E[spend] = ₹175 | sd = ₹455
expected profit per customer from the campaign: ₹-16 → do not run it
```

Expected value drives decisions: pricing, insurance, campaign ROI, and the "expected loss" in credit risk (PD × LGD × EAD).

Discrete distributions



Binomial (number of conversions out of 50 visitors, p = 0.1) and Poisson (support tickets per hour, mean 4).

Distribution	Models	Parameters	Mean	Variance
Bernoulli	One yes/no trial (did this user convert?)	p	p	p(1 − p)
Binomial	Number of successes in n trials	n, p	np	np(1 − p)
Poisson	Count of events in an interval (calls/hour, defects/batch)	λ	λ	λ
Geometric	Trials until first success	p	1/p	(1 − p)/p²

Python

```
b = stats.binom(n=50, p=0.10)
print(f"P(exactly 5) = {b.pmf(5):.3f} | P(at most 2) = {b.cdf(2):.3f} | P(10 or more) =
{b.sf(9):.4f} | mean {b.mean()} sd {b.std():.2f}")
po = stats.poisson(mu=4)
print(f"tickets/hour: P(0) = {po.pmf(0):.3f} | P(more than 8) = {po.sf(8):.4f} | 95th
percentile = {po.ppf(0.95):.0f}")
ab = pd.read_csv("ab_test.csv"); ctrl = ab[ab.group == "A_control"]
print(f"A/B control: {ctrl.converted.sum()} conversions from {len(ctrl)} users, p̂ =
{ctrl.converted.mean():.3f}")
```

OUTPUT

```
P(exactly 5) = 0.185 | P(at most 2) = 0.112 | P(10 or more) = 0.0245 | mean 5.0 sd
2.12
tickets/hour: P(0) = 0.018 | P(more than 8) = 0.0214 | 95th percentile = 8
A/B control: 198 conversions from 2000 users, p̂ = 0.099
```

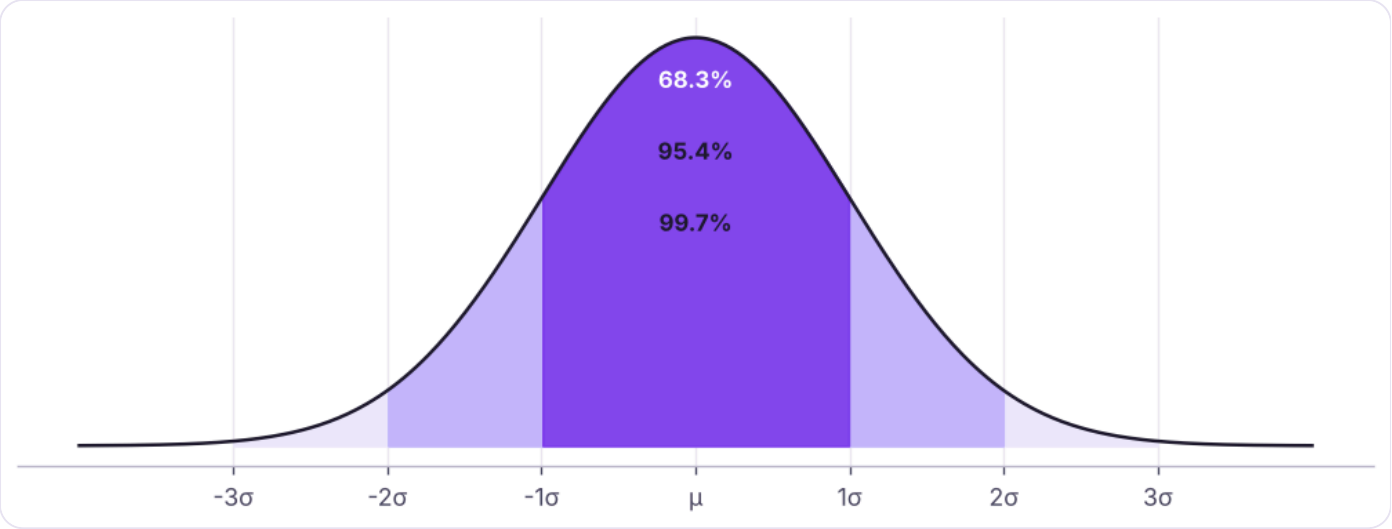
scipy.stats methods: `pmf` / `pdf` (probability or density at a point), `cdf` (P(X ≤ x)), `sf` (P(X > x) = 1 − cdf), `ppf` (inverse cdf: percentiles), `rvs` (random samples).

Continuous distributions

For continuous variables, probability is the **area under the density curve**; P(X = exact value) is 0.

The normal distribution

Bell-shaped, symmetric, described by mean μ and standard deviation σ. Heights, measurement errors and, crucially, **sample means** (next lesson) are approximately normal.



The empirical rule: about 68%, 95% and 99.7% of a normal distribution lie within 1, 2 and 3 standard deviations of the mean.

Python

```
d = pd.read_csv("delivery_times.csv"); pune = d[d.city == "Pune"].delivery_minutes
mu, sd = pune.mean(), pune.std()
nd = stats.norm(mu, sd)
print(f"Pune delivery: mean {mu:.1f} min, sd {sd:.1f}")
print(f"P(late, > 40 min) model {nd.sf(40):.1%} vs data {(pune > 40).mean():.1%}")
print(f"promise time covering 95% of orders: {nd.ppf(0.95):.1f} min | within ±1 sd: {((pune - mu).abs() ⩽ sd).mean():.0%} of orders")
print("68-95-99.7 check:", [round(float(stats.norm.cdf(k) - stats.norm.cdf(-k)), 4) for k in (1, 2, 3)], "| z for 95% two-sided:", round(stats.norm.ppf(0.975), 3))
```

OUTPUT

```
Pune delivery: mean 31.7 min, sd 6.4
P(late, > 40 min) model 9.7% vs data 13.3%
```

```
promise time covering 95% of orders: 42.2 min | within ±1 sd: 68% of orders
68-95-99.7 check: [0.6827, 0.9545, 0.9973] | z for 95% two-sided: 1.96
```

The **standard normal** has μ = 0, σ = 1; any normal value converts with z = (x − μ)/σ. The number **1.96** (95% two-sided) appears in confidence intervals and tests.

Other continuous distributions

Distribution	Models	Example
Uniform(a, b)	Every value equally likely	Random number generators, A/B bucket assignment
Exponential(λ)	Waiting time between Poisson events	Time between support calls
Log-normal	Positive, right-skewed values	Income, house prices, order values
t, χ^2, F	Sampling distributions for tests	t-tests, chi-square tests, ANOVA

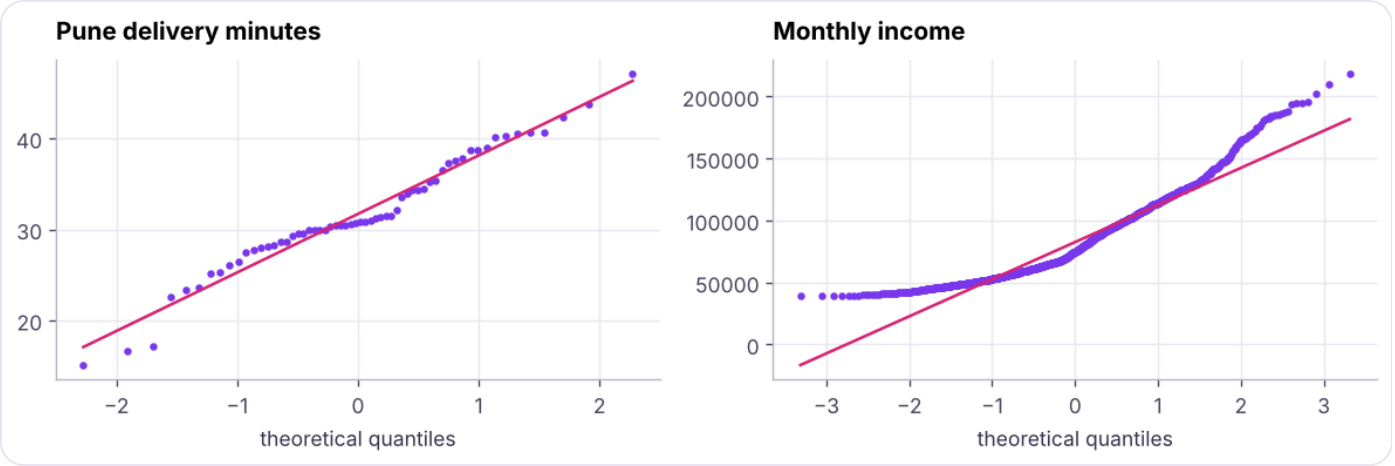
Python

```python
wait = stats.expon(scale=1 / 4)                    # 4 calls per hour → mean wait 15 minutes
print(f"P(no call for 30+ minutes) = {wait.sf(0.5):.3f} | memoryless: P(wait > 45 | already 15) = {wait.sf(0.75) / wait.sf(0.25):.3f}")
```

OUTPUT

```
P(no call for 30+ minutes) = 0.135 | memoryless: P(wait > 45 | already 15) = 0.135
```

Is it normal? Q-Q plots and tests



Q-Q plots compare data quantiles with normal quantiles: points on the line mean "approximately normal". Pune delivery times look normal; HR income is right-skewed.

Python

```
for name, data in [("delivery (Pune)", pune), ("income", hr.monthly_income), ("log income", np.log(hr.monthly_income))]:
    print(f"{name:16s} Shapiro-Wilk p = {stats.shapiro(data.sample(min(len(data), 500), random_state=0)).pvalue:.4f} | skew {stats.skew(data):.2f}")
```

OUTPUT

```
delivery (Pune)  Shapiro-Wilk p = 0.1629 | skew -0.15
income           Shapiro-Wilk p = 0.0000 | skew 1.02
log income       Shapiro-Wilk p = 0.0000 | skew 0.28
```

A small p-value (< 0.05) suggests the data are **not** normal. With large samples, tiny departures become "significant", so rely on plots and skewness too.

Summary

KEY TAKEAWAYS

- Probability rules: complement, addition, multiplication (independence), conditional P(A | B) ≠ P(B | A).
- Bayes' theorem updates beliefs; low base rates make most alerts false positives.
- Expected value = probability-weighted average, the basis of ROI and risk decisions.
- Binomial (successes in n trials), Poisson (counts per interval), normal (measurements, sample means), exponential (waiting times); use `scipy.stats` `pmf/pdf`, `cdf`, `sf`, `ppf`, `rvs`.
- Empirical rule 68-95-99.7; z = (x − μ)/σ; check normality with Q-Q plots and skewness.

Quick quiz

6 questions · tap an answer to check it

1. A fraud test is 99% sensitive with a 2% false-positive rate, and fraud is 0.5% of payments. Roughly what share of flagged payments are fraud?

A 99%

B 50%

C About 20%

D 2%

Answer: C. The low base rate means most flags are genuine payments.

2. Which distribution models the number of support calls per hour?

A Normal

B Poisson

C Uniform

D Binomial with n = 1

Answer: B. Counts of events in a fixed interval.

3. For a binomial with n = 50 and p = 0.1, what is the mean?

A 0.1

B 5

C 50

D 4.5

Answer: B. Mean = np.

4. About what share of a normal distribution lies within ±2 standard deviations?

A 68%

B 95%

C 99.7%

D 50%

Answer: B. The empirical rule.

5. Which scipy method gives P(X > x)?

A pdf

B cdf

C sf

D ppf

Answer: C. sf = survival function = 1 − cdf.

6. P(churn | month-to-month) equals P(month-to-month | churn).

A True

B False

Answer: B. They condition on different groups.

EXERCISE **Practice**

(1) Probability of at least one conversion from 20 visitors if p = 0.08. (2) A call centre averages 6 calls per 10 minutes; probability of more than 10 calls in 10 minutes. (3) Hyderabad delivery times: estimate P(> 45 min) with a normal model and compare with the data. (4) Simulate the Bayes fraud example with fraud at 2% instead of 0.5%. What changes?

Show solution (worked answer)

Python

```
print(1 - 0.92 ** 20, stats.binom.sf(0, 20, 0.08))
print(stats.poisson.sf(10, 6))
hyd = d[d.city == "Hyderabad"].delivery_minutes; print(stats.norm(hyd.mean(),
hyd.std()).sf(45), (hyd > 45).mean())
p_fraud = 0.02; print(sens * p_fraud / (sens * p_fraud + fpr * (1 - p_fraud)))     #
about 50%: the base rate matters
```

INTERVIEW CORNER

Questions you may be asked

Q: Explain Bayes' theorem with an example. It updates a prior probability with new evidence. For a rare disease with an accurate test, a positive result can still be more likely a false positive because healthy people vastly outnumber sick ones.

Q: When would you use a Poisson distribution? For counts of independent events in a fixed time or space with a constant average rate, such as orders per minute, defects per batch or calls per hour.

Q: What is the difference between pdf and cdf? The pdf gives density at a point (probability for discrete pmf); the cdf gives the cumulative probability $P(X \le x)$.

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